

Supermarket Spoilage & Margin Analysis

Business Memo | Prepared by Ahmad Abdulrahim | August 2026

Business Question

Which vegetable categories have the highest loss (spoilage) rates relative to their profit margin, and where should the store adjust order quantities, pricing, or shelf life management to reduce waste-driven losses?

Executive Summary

Flower/Leaf Vegetables is the store's highest-priority category for spoilage-related loss. It shows the highest loss-adjusted margin erosion of any category (12.73%, meaning spoilage wipes out roughly one-eighth of its potential profit) and, more critically, the highest estimated revenue lost to spoilage store-wide at approximately ¥133,328 — more than double the next-highest category. A small set of items within this category — Chuncai, Dongmenkou Xiaobaicai, and Foreign Garland Chrysanthemum — are responsible for the steepest individual erosion and are the most immediate candidates for intervention. Separately, two items were found selling below wholesale cost entirely independent of spoilage, representing a guaranteed loss on every sale that requires its own pricing review.

Finding 1: Flower/Leaf Vegetables Drives the Largest Spoilage-Related Loss

Across all vegetable categories, Flower/Leaf Vegetables shows the highest loss-adjusted margin erosion at 12.73% — the largest share of potential profit wiped out by spoilage of any category analyzed. It also accounts for the highest estimated revenue lost to spoilage store-wide.

Category	Avg. Loss Rate	Margin Erosion	Est. Revenue Lost (RMB)
Flower/Leaf Vegetables	12.73%	12.73%	133,327.63
Cabbage	10.67%	10.66%	39,472.15
Edible Mushroom	9.01%	9.01%	56,816.09
Aquatic Tuberous Vegetables	8.91%	8.91%	33,034.76
Capsicum	7.76%	7.76%	61,083.05
Solanum	6.41%	6.41%	12,281.52

Note: on an absolute (RMB/kg) basis rather than a percentage basis, Edible Mushroom shows marginally higher erosion per kilogram, owing to its higher average margin. Flower/Leaf Vegetables was prioritized here as it leads on both the relative erosion measure and total estimated revenue lost — the more decision-relevant combination for prioritizing action.

Finding 2: A Small Set of Items Drives Most of the Category's Loss

Drilling into Flower/Leaf Vegetables specifically, loss is concentrated in a handful of items rather than spread evenly across the category. The following items show the steepest margin erosion and are the clearest candidates for reduced order volume, shorter reorder cycles, or improved storage handling.

Item	Gross Margin (RMB/kg)	Loss-Adjusted Margin	Margin Erosion
Chuncaï	3.03	2.15	29.03%
Dongmenkou Xiaobaicai	2.13	1.54	27.84%
Foreign Garland Chrysanthemum	6.16	4.55	26.16%
Kuaicai	3.42	2.72	20.38%
Purple Beicai	8.31	6.68	19.58%

For context store-wide: the single largest individual erosion outlier overall is High Melon (Aquatic Tuberous Vegetables, 29.25% erosion) — a reminder that high-erosion items exist outside the worst-performing category too, and are worth monitoring independently of category averages.

Finding 3: Two Items Are Sold Below Wholesale Cost, Independent of Spoilage

Two items — Hongshan Gift Box and Hongshan Shoutidai — show negative gross margin, meaning the store loses money on every unit sold before spoilage is even considered. This is a pricing or data-entry issue, not an inventory management issue, and is not addressed by any of the loss-reduction recommendations below.

Item	Avg. Sale Price (RMB/kg)	Avg. Wholesale Cost (RMB/kg)	Gross Margin (RMB/kg)	Transactions
Hongshan Gift Box	16.27	141.00	-124.73	3
Hongshan Shoutidai	14.83	73.50	-58.67	3

Both items have low transaction volume (3 each), so they do not materially distort category-level averages elsewhere in this analysis — but the per-unit loss is real and consistent, and is worth flagging on its own merits regardless of volume.

Recommendations

1. **Prioritize Flower/Leaf Vegetables for loss-reduction efforts.** It leads all categories in both relative margin erosion and total estimated revenue lost, making it the highest-value target for intervention.
2. **Reduce order volume or shorten reorder cycles for Chuncai, Dongmenkou Xiaobaicai, and Foreign Garland Chrysanthemum specifically,** rather than adjusting ordering for the category as a whole — these three items account for the steepest erosion within the category.
3. **Review storage and shelf-life handling for High Melon** (Aquatic Tuberous Vegetables), the single highest-erosion item store-wide outside the priority category.

4. **Conduct an immediate pricing review for Hongshan Gift Box and Hongshan Shoutidai.** Both are sold below wholesale cost on every transaction — this is a guaranteed loss unrelated to spoilage and should be corrected independently of any inventory-management changes.

Methodology & Limitations

- Margin is calculated using the wholesale price recorded on the same date as each sale, as a proxy for cost basis. This does not account for inventory holding time, but is a reasonable approximation given the fast turnover typical of fresh produce.
- Loss-adjusted margin = gross margin $\times$ (1 – loss rate), representing estimated real profit after accounting for spoilage.
- Two items (Haixian Mushroom variants) were excluded from all loss-rate-based rankings and aggregations due to an implausible 0% loss rate at very high transaction volume (709 and 1,300 transactions respectively) — likely reflecting missing or unrecorded data rather than genuine zero spoilage.
- “Broccoli” had inconsistent category classification across two SKUs in the source data; both were standardized to “Cabbage” for consistent category-level aggregation.
- Loss Rate (%) is treated as a given, pre-calculated input from the source data. It cannot be independently verified or recalculated, as no purchase-quantity data is available — only sale quantities and wholesale prices.
- Full data cleaning steps, additional data quality checks, and SQL queries underlying every figure in this memo are documented in the accompanying analysis notebook.

Data source: Supermarket Sales Data (Kaggle, yapwh1208) — vegetable sales, wholesale pricing, and loss rate data from a Chinese supermarket, July 2020 – June 2023. Analysis performed using SQL (DuckDB) and Python (pandas, matplotlib, seaborn).